

CUSTOMER CHURN INTELLIGENCE PLATFORM

AI-Powered Customer Retention & Predictive Analytics

PROJECT REPORT

Submitted By:

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Project Title:

Customer Churn Intelligence Platform: AI-Powered Customer Retention & Predictive Analytics

ABSTRACT

Customer churn is one of the most significant challenges faced by subscription-based businesses. Losing existing customers not only affects revenue but also increases customer acquisition costs and impacts long-term profitability. The objective of this project was to develop an end-to-end Customer Churn Intelligence Platform capable of identifying customers at risk of leaving and transforming predictive outcomes into actionable business insights.

The project involved data cleaning, exploratory data analysis (EDA), machine learning model development, deployment of a live prediction application, and the creation of an interactive three-page Power BI dashboard. A Random Forest classification model was developed to predict customer churn and deployed through a Streamlit application, enabling real-time prediction capabilities.

The final solution enables organizations to monitor customer health, identify churn drivers, segment customers based on risk levels, and proactively design retention strategies using predictive analytics.

1. INTRODUCTION

Customer churn refers to the loss of customers over a specific period. Understanding why customers leave and identifying those likely to churn are essential for organizations operating in highly competitive markets.

Traditional reporting methods focus primarily on historical trends and often fail to provide future-oriented insights. Predictive analytics addresses this limitation by enabling organizations to anticipate customer behaviour before churn occurs.

This project integrates machine learning, interactive dashboards, and live deployment technologies to transform raw customer data into meaningful business intelligence capable of supporting proactive decision-making.

2. OBJECTIVES

The objectives of this project are:

- To understand customer churn behaviour.
 - To clean and preprocess customer data.
 - To perform exploratory data analysis using Python.
 - To identify major churn drivers.
 - To develop machine learning models for churn prediction.
 - To evaluate model performance using standard classification metrics.
 - To deploy a live prediction application using Streamlit.
 - To develop interactive Power BI dashboards.
 - To segment customers based on predicted risk levels.
 - To provide actionable business recommendations.
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3. TOOLS AND TECHNOLOGIES USED

Programming and Development Tools

- Python
 - Power BI
 - Streamlit
 - Jupyter Notebook
 - Visual Studio Code
 - Git
 - GitHub
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Python Libraries

- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-Learn
 - Joblib
 - Plotly
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4. DATASET INFORMATION

Dataset Name:

IBM Telco Customer Churn Dataset

Target Variable:

Churn

Final Dataset Used:

Telco_Customer_Predictions.csv

The dataset contains customer demographic, service usage, billing, and machine learning prediction information.

Key Feature Categories:

- Customer Information
- Service Information
- Billing Information
- Contract Information
- Churn Outcomes
- Prediction Outputs

- Churn Probabilities

- Risk Categories

5. METHODOLOGY

The project followed an end-to-end analytical workflow consisting of six phases.

Phase 1: Data Understanding and Cleaning

Activities Performed

- Imported the dataset.
- Examined dataset structure.
- Identified variable types.
- Verified missing values.
- Checked duplicate records.
- Converted TotalCharges into numerical format.
- Handled hidden missing values.
- Generated a cleaned dataset.

Outcome

A clean and analysis-ready dataset was prepared for further processing.

Phase 2: Exploratory Data Analysis (EDA)

Analyses Conducted

- Churn Distribution Analysis
- Gender Analysis
- Senior Citizen Analysis
- Contract Analysis
- Internet Service Analysis

- Payment Method Analysis
- Monthly Charges Analysis
- Tenure Analysis
- Service Subscription Analysis
- Correlation Analysis

Outcome

EDA revealed several customer segments and factors strongly associated with churn behaviour.

Phase 3: Feature Engineering

Activities Performed

- Encoded categorical variables.
- Performed train-test split.
- Applied feature scaling.
- Prepared model-ready datasets.

Outcome

The dataset was transformed into a suitable format for machine learning model development.

Phase 4: Machine Learning Modeling

Models Evaluated

- Logistic Regression
 - Decision Tree
 - Random Forest
 - K-Nearest Neighbors
 - Support Vector Machine
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Best Deployed Model

Random Forest Classifier

Model Performance

Accuracy: 75.20%

Precision: 52.22%

Recall: 78.61%

F1 Score: 62.75%

ROC-AUC Score: 83.60%

Cross Validation ROC-AUC: 84.40%

Optimal Threshold: 0.5905

Outcome

The Random Forest model demonstrated strong predictive capability and was selected for deployment.

Phase 5: Live Prediction Application

Application Features

- Real-time churn prediction.
 - Interactive customer input interface.
 - Churn probability estimation.
 - Risk categorization.
 - Personalized prediction output.
 - Modern Streamlit user interface.
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Deployment Components

- Streamlit
- Random Forest Model

- Scaler
 - Feature Encoder
 - Prediction Engine
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Outcome

A fully functional live prediction application was developed to enable real-time churn assessment.

Phase 6: Power BI Dashboard Development

Three interactive dashboard pages were developed.

Page 1: Customer Health Overview

Purpose:

To answer the question:

"What is happening?"

Visuals:

- KPI Cards
 - Customer Health Distribution
 - Contract Impact Analysis
 - Internet Service and Churn Analysis
 - Interactive Filters
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Page 2: Churn Intelligence

Purpose:

To answer the question:

"Why is it happening?"

Visuals:

- Contract Type vs Churn
 - Internet Service vs Predicted Churn
 - Payment Method vs Churn
 - Customer Driver KPIs
 - Interactive Filters
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Page 3: Predictive Analysis

Purpose:

To answer the question:

"What is likely to happen next?"

Visuals:

- Risk Category Distribution
 - Actual vs Predicted Churn
 - Risk Segmentation Analysis
 - Live Prediction Center
 - Interactive Filters
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6. KEY FINDINGS

The major findings of the project include:

- Month-to-month customers exhibited the highest churn.
- Fiber optic users demonstrated elevated churn tendencies.
- Electronic check customers showed stronger churn behaviour.
- Customers with shorter tenure were more vulnerable to churn.
- High-risk customers represented significant retention opportunities.
- The Random Forest model effectively identified churn-prone customers.

- Predictive analytics enhanced decision-making beyond traditional reporting.
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7. BUSINESS RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

- Monitor churn KPIs regularly.
 - Encourage long-term contract adoption.
 - Develop targeted retention campaigns.
 - Improve customer engagement during early tenure.
 - Focus retention initiatives on high-risk customer segments.
 - Review service offerings for vulnerable groups.
 - Integrate predictive monitoring into operational workflows.
 - Continuously evaluate model performance.
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8. CONCLUSION

This project successfully demonstrated the application of machine learning, business intelligence, and deployment technologies in addressing a real-world customer retention problem.

The developed Customer Churn Intelligence Platform transformed raw customer data into actionable insights through exploratory analysis, predictive modeling, live deployment, and interactive reporting.

The findings highlighted that customer churn is influenced by multiple interconnected factors involving customer behaviour, service usage, contract characteristics, and billing patterns.

The final solution provides organizations with a practical framework to monitor churn trends, identify vulnerable customer groups, predict future churn behaviour, and implement proactive retention strategies.

9. SKILLS DEMONSTRATED

The project demonstrates proficiency in:

- Data Cleaning

- Exploratory Data Analysis
 - Feature Engineering
 - Machine Learning
 - Model Evaluation
 - Predictive Analytics
 - Streamlit Deployment
 - Power BI Dashboard Development
 - DAX Development
 - Executive Reporting
 - Business Storytelling
 - GitHub Project Management
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10. PROJECT OUTCOME

An end-to-end Customer Churn Intelligence solution was successfully developed covering:

- Data Understanding
- Data Cleaning
- Exploratory Data Analysis
- Feature Engineering
- Machine Learning Modeling
- Live Prediction Deployment
- Interactive Power BI Dashboard Development

The completed project provides organizations with an analytical framework to support proactive customer retention and strategic decision-making through predictive intelligence.

Submitted By

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Data Analyst Portfolio Project

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